
D Matrices

Matrices arise in numerous applications, including, but by no means limited to, scientific computing. If you have seen matrices before, much of the material in this appendix will be familiar to you, but some of it might be new. Section D.1 covers basic matrix definitions and operations, and Section D.2 presents some basic matrix properties.

D.1 Matrices and matrix operations

This section reviews some basic concepts of matrix theory and some fundamental properties of matrices.

Matrices and vectors

A *matrix* is a rectangular array of numbers. For example,

$$\begin{aligned} A &= \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \\ &= \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \end{aligned} \tag{D.1}$$

is a 2×3 matrix $A = (a_{ij})$, where for $i = 1, 2$ and $j = 1, 2, 3$, the element of the matrix in row i and column j is denoted by a_{ij} . By convention, uppercase letters denote matrices and corresponding subscripted lowercase letters denote their elements. We denote the set of all $m \times n$ matrices with real-valued entries by $\mathbb{R}^{m \times n}$ and, in general, the set of $m \times n$ matrices with entries drawn from a set S by $S^{m \times n}$.

The **transpose** of a matrix A is the matrix A^T obtained by exchanging the rows and columns of A . For the matrix A of equation (D.1),

$$A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

A **vector** is a one-dimensional array of numbers. For example,

$$x = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$$

is a vector of size 3. We sometimes call a vector of length n an **n -vector**. By convention, lowercase letters denote vectors, and the i th element of a size- n vector x is denoted by x_i , for $i = 1, 2, \dots, n$. We take the standard form of a vector to be as a **column vector** equivalent to an $n \times 1$ matrix, whereas the corresponding **row vector** is obtained by taking the transpose:

$$x^T = (2 \ 3 \ 5).$$

The **unit vector** e_i is the vector whose i th element is 1 and all of whose other elements are 0. Usually, the context makes the size of a unit vector clear.

A **zero matrix** is a matrix all of whose entries are 0. Such a matrix is often denoted 0, since the ambiguity between the number 0 and a matrix of 0s can usually be resolved from context. If a matrix of 0s is intended, then the size of the matrix also needs to be derived from the context.

Square matrices

Square $n \times n$ matrices arise frequently. Several special cases of square matrices are of particular interest:

1. A **diagonal matrix** has $a_{ij} = 0$ whenever $i \neq j$. Because all of the off-diagonal elements are 0, a succinct way to specify the matrix lists only the elements along the diagonal:

$$\text{diag}(a_{11}, a_{22}, \dots, a_{nn}) = \begin{pmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}.$$

2. The $n \times n$ **identity matrix** I_n is a diagonal matrix with 1s along the diagonal:

$$\begin{aligned} I_n &= \text{diag}(1, 1, \dots, 1) \\ &= \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}. \end{aligned}$$

When I appears without a subscript, its size derives from the context. The i th column of an identity matrix is the unit vector e_i .

3. A **tridiagonal matrix** T is one for which $t_{ij} = 0$ if $|i - j| > 1$. Nonzero entries appear only on the main diagonal, immediately above the main diagonal ($t_{i,i+1}$ for $i = 1, 2, \dots, n - 1$), or immediately below the main diagonal ($t_{i+1,i}$ for $i = 1, 2, \dots, n - 1$):

$$T = \begin{pmatrix} t_{11} & t_{12} & 0 & 0 & \dots & 0 & 0 & 0 \\ t_{21} & t_{22} & t_{23} & 0 & \dots & 0 & 0 & 0 \\ 0 & t_{32} & t_{33} & t_{34} & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & t_{n-2,n-2} & t_{n-2,n-1} & 0 \\ 0 & 0 & 0 & 0 & \dots & t_{n-1,n-2} & t_{n-1,n-1} & t_{n-1,n} \\ 0 & 0 & 0 & 0 & \dots & 0 & t_{n,n-1} & t_{nn} \end{pmatrix}.$$

4. An **upper-triangular matrix** U is one for which $u_{ij} = 0$ if $i > j$. All entries below the diagonal are 0:

$$U = \begin{pmatrix} u_{11} & u_{12} & \dots & u_{1n} \\ 0 & u_{22} & \dots & u_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & u_{nn} \end{pmatrix}.$$

An upper-triangular matrix is **unit upper-triangular** if it has all 1s along the diagonal.

5. A **lower-triangular matrix** L is one for which $l_{ij} = 0$ if $i < j$. All entries above the diagonal are 0:

$$L = \begin{pmatrix} l_{11} & 0 & \dots & 0 \\ l_{21} & l_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ l_{n1} & l_{n2} & \dots & l_{nn} \end{pmatrix}.$$

A lower-triangular matrix is **unit lower-triangular** if it has all 1s along the diagonal.

6. A **permutation matrix** P has exactly one 1 in each row or column, and 0s elsewhere. An example of a permutation matrix is

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

Such a matrix is called a permutation matrix because multiplying a vector x by a permutation matrix has the effect of permuting (rearranging) the elements of x . Exercise D.1-4 explores additional properties of permutation matrices.

7. A **symmetric matrix** A satisfies the condition $A = A^T$. For example,

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 6 & 4 \\ 3 & 4 & 5 \end{pmatrix}$$

is a symmetric matrix.

Basic matrix operations

The elements of a matrix or vector are **scalar numbers** from a number system, such as the real numbers, the complex numbers, or integers modulo a prime. The number system defines how to add and multiply scalars. These definitions extend to encompass addition and multiplication of matrices.

We define **matrix addition** as follows. If $A = (a_{ij})$ and $B = (b_{ij})$ are $m \times n$ matrices, then their matrix sum $C = (c_{ij}) = A + B$ is the $m \times n$ matrix

defined by

$$c_{ij} = a_{ij} + b_{ij}$$

for $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$. That is, matrix addition is performed componentwise. A zero matrix is the identity for matrix addition:

$$A + 0 = A = 0 + A.$$

If λ is a scalar number and $A = (a_{ij})$ is a matrix, then $\lambda A = (\lambda a_{ij})$ is the *scalar multiple* of A obtained by multiplying each of its elements by λ . As a special case, we define the *negative* of a matrix $A = (a_{ij})$ to be $-1 \cdot A = -A$, so that the ij th entry of $-A$ is $-a_{ij}$. Thus,

$$A + (-A) = 0 = (-A) + A.$$

The negative of a matrix defines *matrix subtraction*: $A - B = A + (-B)$.

We define *matrix multiplication* as follows. Start with two matrices A and B that are *compatible* in the sense that the number of columns of A equals the number of rows of B . (In general, an expression containing a matrix product AB is always assumed to imply that matrices A and B are compatible.) If $A = (a_{ik})$ is a $p \times q$ matrix and $B = (b_{kj})$ is a $q \times r$ matrix, then their matrix product $C = AB$ is the $p \times r$ matrix $C = (c_{ij})$, where

$$c_{ij} = \sum_{k=1}^q a_{ik} b_{kj} \tag{D.2}$$

for $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, p$. The procedure RECTANGULAR-MATRIX-MULTIPLY on page 374 implements matrix multiplication in the straightforward manner based on equation (D.2), assuming that C is initialized to 0, using pqr multiplications and $p(q-1)r$ additions for a running time of $\Theta(pqr)$. If the matrices are $n \times n$ square matrices, so that $n = p = q = r$, the pseudocode reduces to MATRIX-MULTIPLY on page 81, whose running time is $\Theta(n^3)$. (Section 4.2 describes an asymptotically faster $\Theta(n^{\lg 7})$ -time algorithm due to V. Strassen.)

Matrices have many (but not all) of the algebraic properties typical of numbers. Identity matrices are identities for matrix multiplication:

$$I_m A = A I_n = A$$

for any $m \times n$ matrix A . Multiplying by a zero matrix gives a zero matrix:

$$A \cdot 0 = 0.$$

Matrix multiplication is associative:

$$A(BC) = (AB)C$$

for compatible matrices A , B , and C . Matrix multiplication distributes over addition:

$$A(B + C) = AB + AC,$$

$$(B + C)D = BD + CD.$$

For $n > 1$, multiplication of $n \times n$ matrices is not commutative. For example, if $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, then $AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $BA = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.

We define matrix-vector products or vector-vector products as if the vector were the equivalent $n \times 1$ matrix (or a $1 \times n$ matrix, in the case of a row vector). Thus, if A is an $m \times n$ matrix and x is an n -vector, then Ax is an m -vector. If x and y are n -vectors, then

$$x^T y = \sum_{i=1}^n x_i y_i$$

is a scalar number (actually a 1×1 matrix) called the **inner product** of x and y . We also use the notation $\langle x, y \rangle$ to denote $x^T y$. The inner-product operator is commutative: $\langle x, y \rangle = \langle y, x \rangle$. The matrix xy^T is an $n \times n$ matrix Z called the **outer product** of x and y , where $z_{ij} = x_i y_j$. The **(euclidean) norm** $\|x\|$ of an n -vector x is defined by

$$\begin{aligned} \|x\| &= (x_1^2 + x_2^2 + \dots + x_n^2)^{1/2} \\ &= (x^T x)^{1/2}. \end{aligned}$$

Thus, the norm of x is its length in n -dimensional euclidean space. A useful fact, which follows from the equality

$$((ax_1)^2 + (ax_2)^2 + \cdots + (ax_n)^2)^{1/2} = |a| (x_1^2 + x_2^2 + \cdots + x_n^2)^{1/2}$$

is that for any real number a and n -vector x ,

$$\|ax\| = |a| \|x\| . \quad (\text{D.3})$$

Exercises

D.1-1

Show that if A and B are symmetric $n \times n$ matrices, then so are $A + B$ and $A - B$.

D.1-2

Prove that $(AB)^T = B^T A^T$ and that $A^T A$ is always a symmetric matrix.

D.1-3

Prove that the product of two lower-triangular matrices is lower-triangular.

D.1-4

Prove that if P is an $n \times n$ permutation matrix and A is an $n \times n$ matrix, then the matrix product PA is A with its rows permuted, and the matrix product AP is A with its columns permuted. Prove that the product of two permutation matrices is a permutation matrix.

D.2 Basic matrix properties

We now define some basic properties pertaining to matrices: inverses, linear dependence and independence, rank, and determinants. We also define the class of positive-definite matrices.

Matrix inverses, ranks, and determinants